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Class 7 Math Notes In English Chapter – 10 Practical Geometry

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1. If two lines are cut by a transversal such that their alternate angles are equal, then the lines are parallel. This concept is used to construct a line parallel to a given line l passing through a point not on the line l.

2. A triangle is said to be
    a. an equilateral triangle, if all of its sides are equal.
    b. an isosceles triangle, if any two of its sides are equal.
    c. a scalene triangle, if all of its sides are of different length.

3. A triangle is said to be
    a. an acute angled triangle, if each one of its angles measure less than90^o.
    b. a right-angled triangle, if any one of its angle's measures 90°.
    c. an obtuse angled triangle, if any one of its angles measure more than 90°.

4. A triangle can be constructed if the lengths of its three sides are given.

5. A triangle can be constructed if the length of its two sides and the included angle are given.

6. A triangle can be constructed if its two angles and the length of the included side are given.

7. A right angled triangle can be constructed if we are given the hypotenuse and the length of one of its legs.

Lines That Don't Meet

Method of construction of a line parallel to a given line, using only a sheet of paper

➥ Take a piece of paper.

➥ Fold it in half and unfold the line l. Mark a point A on paper outside l.

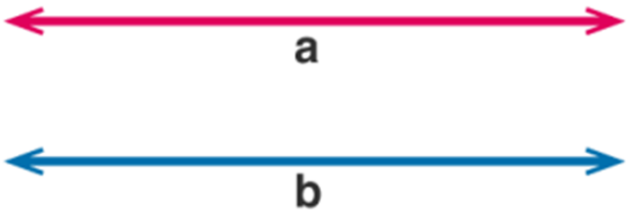
➥ Fold the paper perpendicular to the line such that this perpendicular passes through A. Name the perpendicular AN.

➥ Make a fold perpendicular to AN through point A. Name the new perpendicular line as m.

➥ Now, l || m.

Parallel Lines

Have you ever observed the railway tracks or the opposite edges of a door? One of the interesting facts about these is that they are examples of parallel lines. Now the question arises what exactly are parallel lines? The lines which do not have a common meeting point in the same plane, however far they are extended are known as parallel lines. The following figure represents parallel lines:

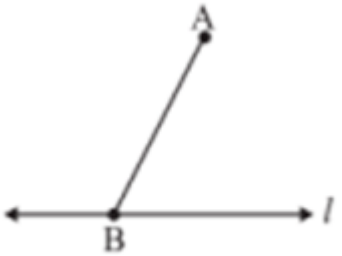


Two lines are said to be parallel in a plane if they do not intersect if extended till infinite in both the directions. The distance between two lines is similar throughout the whole length. The symbol to denote the parallel lines is ||. Suppose the two lines 'a' and 'b' are parallel to each other, we can express their parallel nature using a symbol by a||b and read as 'a is parallel to b'.

Steps of construction of a line parallel to a given line

➥ Take a line l and a point A outside l.

➥ Take any point B on l and join it to A.SSS



➥ With B as the centre and a convenient radius, cut an arc on l at C and BA at D.

➥ With A as the centre and same radius as in Step 3, cut an arc EF to cut AB at G.

➥ Measure the arc length CD by placing pointed tip of the compass at C and pencil tip opening at D.

➥ With this opening, keep G as centre and draw an arc to cut arc EF at H

➡ Join AH to draw a line m

Slope

In Mathematics, a slope of a line is the change in y coordinate with respect to the change in x coordinate.

The net change in y-coordinate is represented by Δy and the net change in x-coordinate is represented by Δx.

Hence, the change in y-coordinate with respect to the change in x-coordinate is given by, m = change in y/change in x = $\Delta y/\Delta x$

Where "m" is the slope of a line.

The slope of the line can also be represented by

$\tan \theta = \Delta y/\Delta x$

So, $\tan \theta$ to be the slope of a line.

Generally, the slope of a line gives the measure of its steepness and direction. The slope of a straight line between two points says (x_1, y_1) and $(x_2, y2)$ can be easily determined by finding the difference between the coordinates of the points. The slope is usually represented by the letter 'm'.

Slope Formula

If P (x_1, y_1) and Q (x_2, y_2) are the two points on a straight line, then the slope formula is given by:

Slope, m = Change in y-coordinates/Change in x-coordinates

$m = (y_2 - y_1)/ (x_2 - x_1)$

Therefore, based on the above formula, we can easily calculate the slope of a line between two points.

In other term, the slope of a line between two points is also said to be the rise of the line from one point to another (along y-axis) over the run (along x-axis). Therefore,

Slope, m = Rise/Run

Slope of a Line Equation

The equation for the slope of a line and the points also called point slope form of equation of a straight line is given by:

$y - y_1 = m (x - x_1)$

Whereas the slope-intercept form the equation of the line is given by:

$y = mx + b$

Where b is the y-intercept.

How to Find Slope of a Line on a Graph?

In the given figure, if the angle of inclination of the given line with the x-axis is θ then, the slope of the line is given by tan θ. Hence, there is a relation between the lines and angles. In this article, you will learn various formulas related to the angles and lines.

The slope of a line is given as $m = \tan\theta$. If two points A (x_1, y_1) and B (x_2, y_2) lie on the line with $(x_1 \neq x_2)$ then the slope of the line AB is given as:

$$m = \tan\theta = (y_2 - y_1)/(x_2 - x_1)$$

Where θ is the angle which the line AB makes with the positive direction of the x-axis. θ lies between 0° and 180°.

It must be noted that $\theta = 90°$ is only possible when the line is parallel to y-axis i.e. at $x_1 = x_2$ at this particular angle the slope of the line is undefined.

Conditions for perpendicular, parallelism, and collinearity of straight lines are given below:

Slope for Parallel Lines

Consider two parallel lines given by l1 and l2 with inclinations α and β respectively. For two lines to be parallel their inclination must also be equal i.e. $\alpha = \beta$. This results in the fact that $\tan\alpha = \tan\beta$. Hence, the condition for two lines with inclinations α, β to be parallel is $\tan\alpha = \tan\beta$.

Therefore, if the slopes of two lines on the Cartesian plane are equal, then the lines are parallel to each other.

Thus, if two lines are parallel then, $m_1 = m_2$.

Generalizing this for n lines, they are parallel only when the slopes of all the lines are equal.

If the equation of the two lines are given as $ax + by + c = 0$ and $a' x + b' y + c' = 0$, then they are parallel when $ab' = a'b$.

Let's Build Triangles

Classification of triangles based on sides and angles

Triangles can be classified based on their:

Sides:

Equilateral triangle: All three sides are equal in measure.

Isosceles triangle: Two sides have equal measure.

Scalene triangle: All three sides have different measures.

Angles:

Acute triangle: All angles measure less than 90^0.

Obtuse triangle: One angle is greater than 90^0.

Right triangle: One angle is 90^0.

Important properties of triangles

- ➡ The exterior angle is equal to the sum of interior opposite angles.
- ➡ The sum of all interior angles is 180°
- ➡ Sum of the lengths of any two sides is greater than the length of the third side.
- ➡ Pythagoras theorem: In any right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Triangles can be constructed if any of the following measurements are given

- ➥ Three sides.
- ➥ Two sides and an angle between them.
- ➥ Two angles and a side between them.
- ➥ The hypotenuse and a leg in case of a right-angled triangle.

Construction of triangles given a criterion are listed below:

Construction of a triangle with SSS criterion.

Construct a triangle ABC, given that AB = 4.5 cm, BC = 5 cm and AC = 6 cm. Steps:

➥ Make a rough sketch for your reference

➥ Draw a line segment BC = 5 cm

➥ With B as centre, draw an arc of radius 4.5 cm

➥ With C as centre, draw an arc of radius 6 cm and cut the previous arc

➥ Mark the point of intersection of arcs as A. Join AB and AC. ΔABC is now ready

Note: SSS congruency rule: If three sides of one triangle are equal to the corresponding three sides of another triangle, then the two triangles are congruent

Construction of a triangle with SAS criterion

Construct ΔPQR with QR = 7.5 cm, PQ = 5 cm and ∠Q = 600.

Steps:

➥ Make a rough sketch for your reference

➥ Draw a line segment QR = 7.5 cm

➥ At Q, draw QX making 600 with QR

➡ With Q as centre, draw an arc of radius 5 cm. It cuts QX at P.

➡ Join AB. ΔPQR is now ready

Construction of a triangle with ASA criterion

Construct ΔXYZ with $\angle X = 30^0$, $\angle Y = 100^0$and XY = 5.8 cm.

Steps:

➡ Make a rough sketch for your reference

➡ Draw XY = 5.8cm

➥ At X, draw a ray XP making an angle of 300 with AB.

➥ At Y, draw a ray YQ making an angle of 1000 with XY.

➥ The point of intersection of the two rays is Z.

➥ ΔXYZ is now completed

Construction of a triangle with RHS criterion

Construct ΔLMN, where $\angle M = 90^0$, MN = 8cm and LN = 10 cm.

Steps:

➥ Make a rough sketch for your reference

➥ Draw MN = 8 cm

➥ At M, draw MX ⊥ MN.

➥ With N as centre, draw an arc of radius 10 cm to cut MX at L

➥ Join LN.

➥ ΔLMN is now completed

Basics of Practical Geometry

Introduction to Constructions of basic figures

Basic constructions:

- ➥ To draw a line segment of given length
- ➥ a line perpendicular to a given line segment
- ➥ an angle
- ➥ an angle bisector
- ➥ a circle

In geometry, you have already studied basic constructions.

To recap, some of them are:

- ➥ Drawing a line segment of given length
- ➥ Drawing a line perpendicular to a given line segment.
- ➥ Angles
- ➥ Angle bisectors
- ➥ Circles
- ➥ Tools used for simple constructions are ruler, protractor and a compass.

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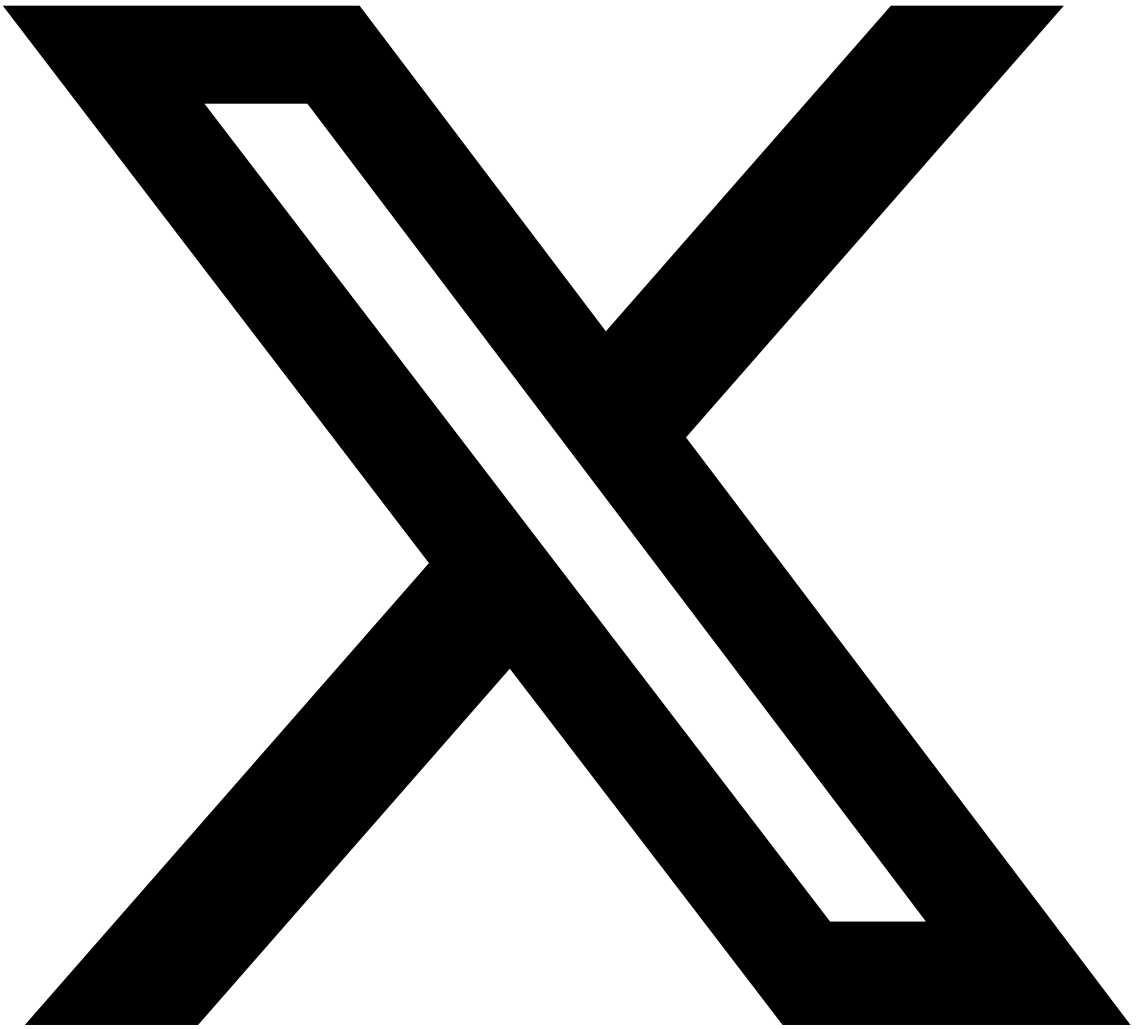

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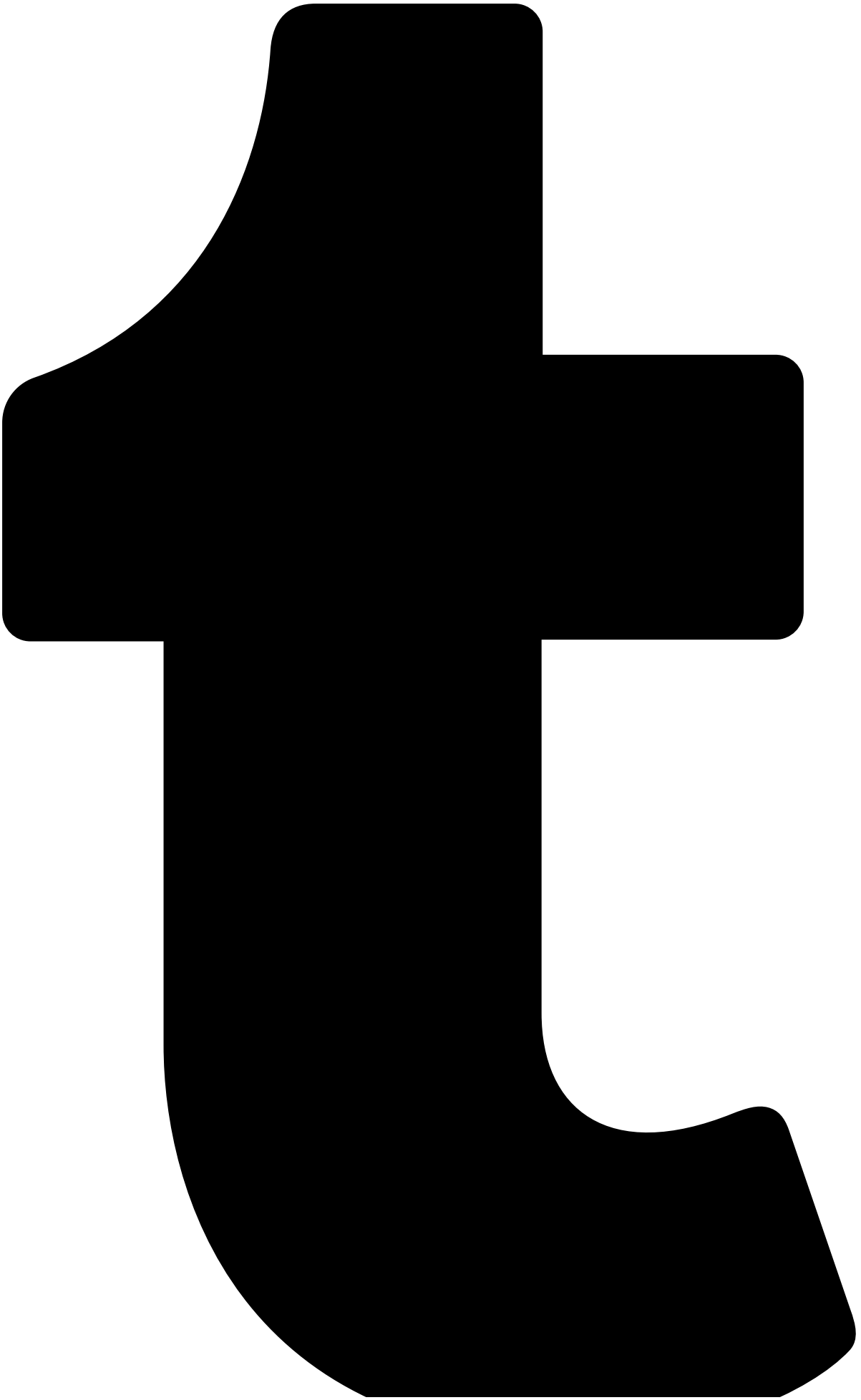

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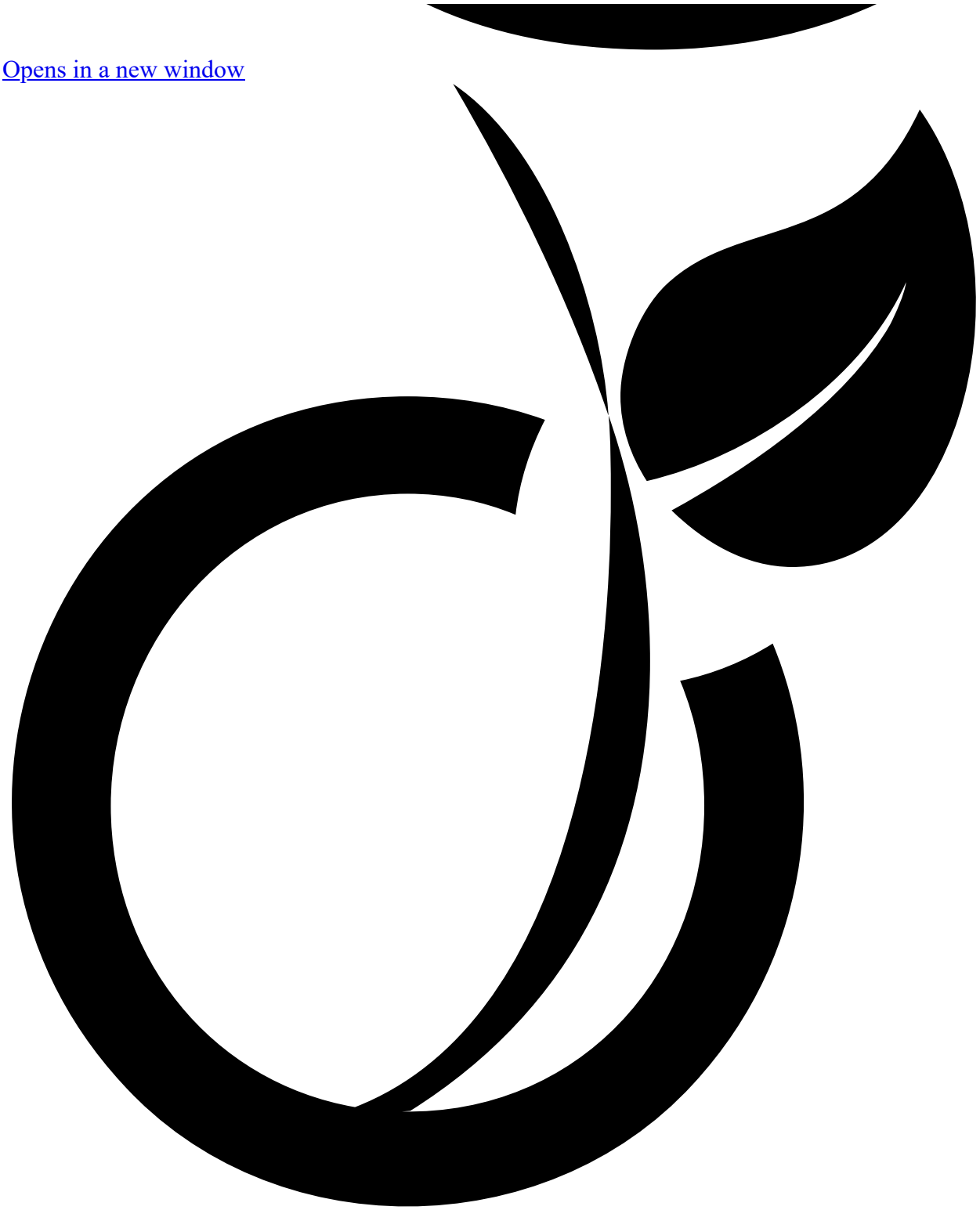

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